

INFO6007

Project Management in IT

Week 04: Cost Management Plan

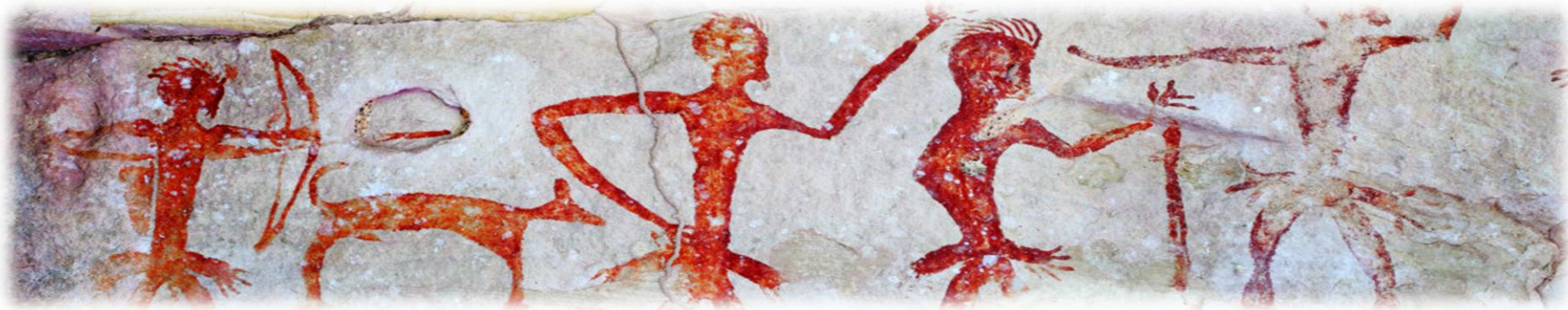
Semester 1, 2026



Acknowledgement of Country

I would like to acknowledge the Traditional Owners of Australia and recognise their continuing connection to land, water and culture. I am currently on the land of the Gadigal people of the Eora nation and pay my respects to their Elders, past, present and emerging.

I further acknowledge the Traditional Owners of the country on which you are on and pay respects to their Elders, past, present and future.



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Agenda

- Introduction
- Project Cost Management

Project Cost Management

Introduction

- What is Project Cost Management?
 - It is the process of **estimating, budgeting, and controlling costs**.
 - Ensures project completion within approved budget.
 - Covers **resources, labor, infrastructure, tools, vendors, risk reserves**.
- The formal definition for Project Cost Management:
 - Project Cost Management includes the processes involved in planning, estimating, budgeting, financing, funding, managing, and controlling costs so that the project can be completed within the approved budget.

Introduction

- A project that is on time but **over budget** is considered a failure by stakeholders.
- IT Cost Management is critical due to:
 - Rapidly changing technology = volatile pricing
 - Hidden costs (integration, security, compliance)
 - Vendor lock-in & license renewals
- Consequences of poor Cost Management:
 - Project cancellation
 - Loss of stakeholder trust
 - Missed opportunities / sunk costs

Types of costs

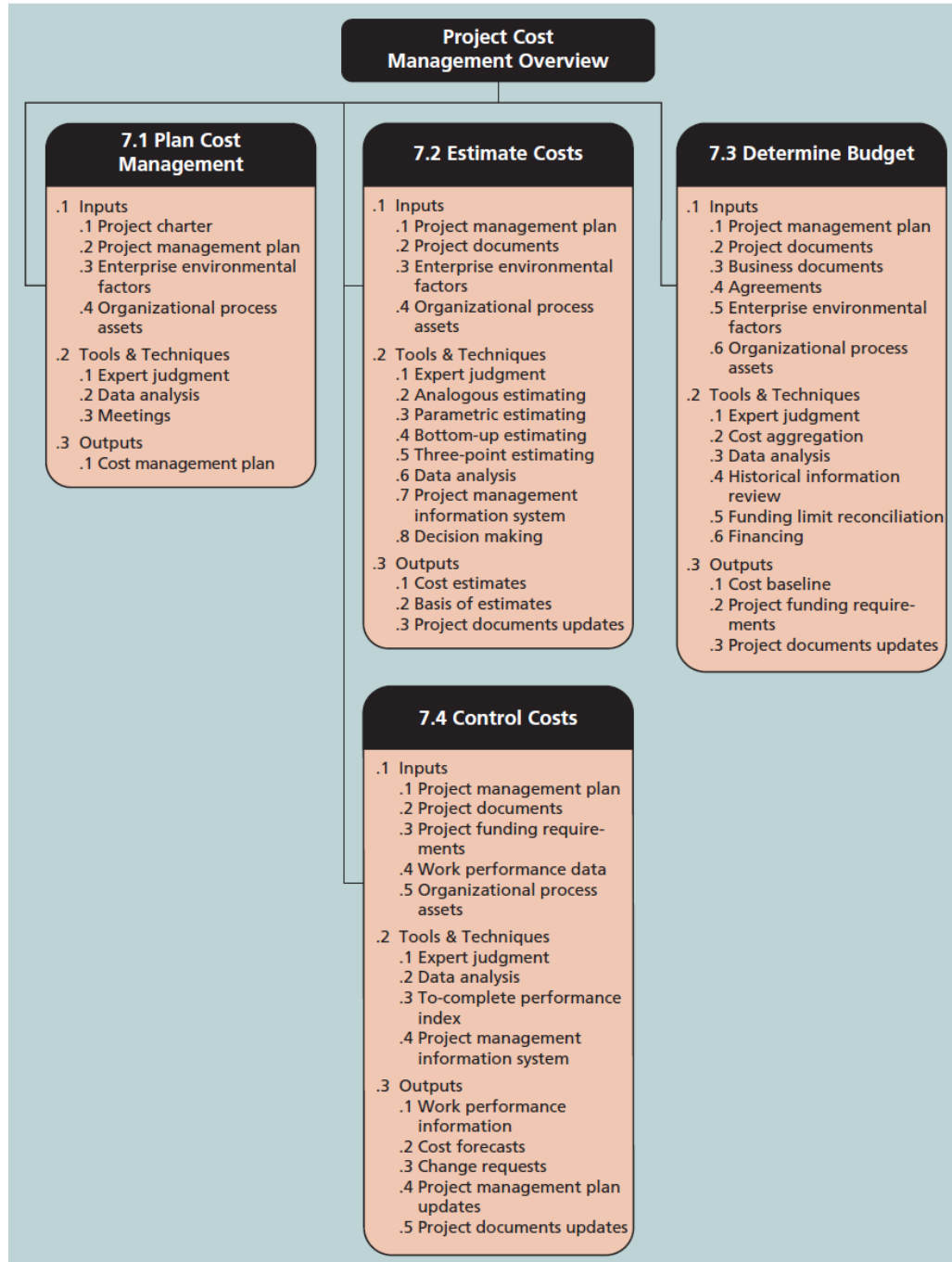
- Cost is a resource sacrificed or foregone to achieve a specific objective, or something given up in exchange.
- Costs are usually measured in monetary units like dollars.
- Types of costs include:
 - **Tangible costs:** Costs that an organisation can easily measure in dollars.
 - **Intangible costs:** Costs that are difficult to measure in monetary terms.
 - **Direct costs:** Costs that are directly related to producing the products and services of the project.
 - **Indirect costs:** Costs that are not directly related to the products or services but indirectly related to performing the project.
 - **Sunk costs:** Money that has been spent in the past when deciding what projects to invest in. Sunk costs are not included in the PMP.

What is a reserve?

- Reserve is the amount of money included in a cost estimate to mitigate the cost risk by allowing for future situations that are difficult to predict.
- There are 2 types of reserves:
 - **Contingency reserves**
 - These reserves allow for future situations that may be partially planned for and are included in the project cost baseline.
 - **Management reserves**
 - These reserves allow for future situations that are unpredictable.

Key components of Cost Management Plan

Component	Description
Planning Cost Management	Determining the policies, procedures, and documentations that will be used for planning, executing, and controlling project cost.
Estimating Costs	Developing an approximation or estimate of the costs of the resources needed to complete a project.
Determining the Budget	Allocating the overall cost estimate to individual work items to establish a baseline for measuring performance.
Controlling Costs	Controlling changes to the project budget.



4 main processes in Project Cost Management

1. Planning Cost Management

- This is the process of defining how project costs will be estimated, budgeted, managed, monitored, and controlled.
- This step provides a **framework** for making cost-related decisions throughout the project lifecycle.

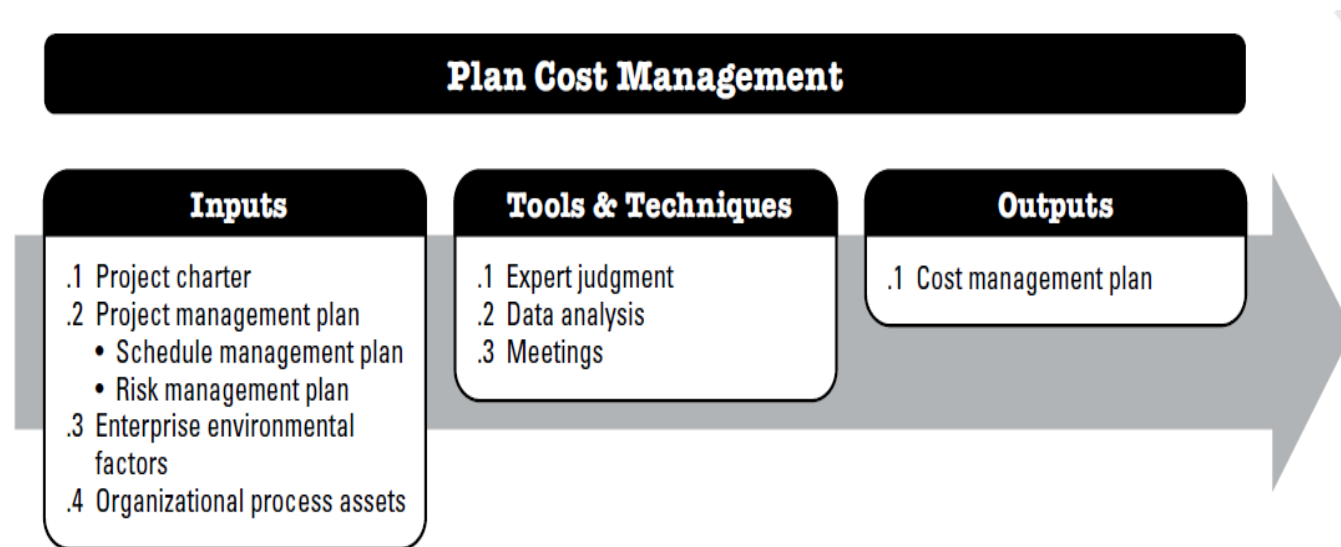


Figure 7-2. Plan Cost Management: Inputs, Tools & Techniques, and Outputs

1. Planning Cost Management

- The purpose of planning cost management:
 - Ensures consistency in cost estimation and control.
 - Provides guidelines for resource allocation and budget approvals.
 - Defines roles & responsibilities (e.g., who approves changes in budget).
 - Aligns with organizational financial policies and compliance standards.
 - Helps prevent budget overruns and scope-related disputes.

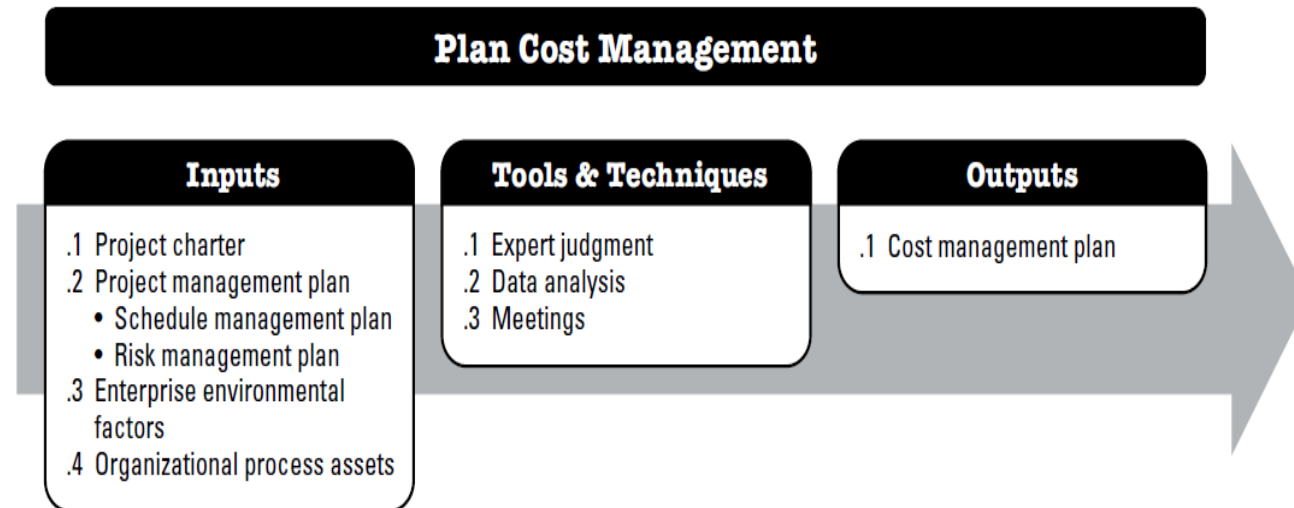


Figure 7-2. Plan Cost Management: Inputs, Tools & Techniques, and Outputs

2. Estimating Costs

- It is the process of developing an approximation of the monetary resources needed to complete project activities.
- The main purpose of this step is to **predict costs** as accurately as possible so stakeholders can allocate resources and approve budgets.
- This step is very important as:
 - IT projects have uncertain requirements → costs are harder to lock in.
 - Rapid tech changes → hardware/software costs fluctuate.
 - Labour-intensive → developer hours often underestimated.
 - Stakeholder decisions (e.g., feature additions) directly impact costs.

Techniques to estimate cost

- Analogous Estimating
 - Based on historical data from similar projects. It is quick but less accurate.
- Parametric Estimating
 - Uses statistical/quantitative models.
- Bottom-Up Estimating
 - Estimate each **work package/task** → roll up to total. This is the most detailed and hence the most accurate technique.
- Three Point Estimating (PERT)
 - Similar to Schedule estimating technique. Accounts for uncertainty.
- Expert Judgement
 - Relying on domain experts or consultants.

Sample Cost Estimate

- Bottom-Up Estimation

Develop a Mobile Banking App

- Login Module: 120 hours × \$60/hr = **\$7,200**
- Payments Module: 300 hours × \$60/hr = **\$18,000**
- Testing: 200 hours × \$50/hr = **\$10,000**
- Cloud Hosting (1 year): **\$12,000**
- Subtotal = **\$47,200**
- Contingency = **\$4,720**

- Three-Point Estimation

Set up cloud infrastructure

- Optimistic (O): **\$8,000**
- Most Likely (M): **\$12,000**
- Pessimistic (P): **\$20,000**
- Estimate = $(8000 + 4(12000) + 20000) / 6$
= **\$12,667.00**

What affects IT Cost Estimation?

- Factors affecting accuracy:
 - Project phase (early vs late).
 - Quality of requirements (clear vs vague).
 - Estimation technique chosen.
 - Availability of historical data.
 - Team expertise.
- Common mistakes in IT Cost estimation:
 - Underestimating integration, testing, and maintenance costs.
 - Ignoring hidden costs (training, security compliance, vendor support).
 - Over-reliance on optimistic timelines.
 - Forgetting currency fluctuations in global projects.

Software Tools for Cost Estimation & Control

- **Project Management Software**

- MS Project, Primavera, Oracle Cost Manager
- Support cost baselines, resource tracking, earned value calculations

- **Agile and Collaboration Tools**

- Jira with add-ons (Tempo, BigPicture) for sprint cost tracking
- Azure DevOps for release budgeting and resource burn monitoring

- **Data Visualization and Dashboards**

- Power BI, Tableau for real-time cost dashboards and variance monitoring
- Enhance communication of cost performance to stakeholders

Benchmarking in Cost Estimation

- What is Benchmarking?
 - Using historical project data or industry averages to guide cost estimation.
- Examples
 - Average cost per migrated user to cloud services (e.g., \$150–\$250 per user)
 - Cost per function point or story point in software development
 - Average sprint cost for an Agile team (e.g., \$40K–\$60K for two weeks)
- Benefits
 - Anchors estimates in realistic ranges
 - Highlights outliers that need review
 - Builds a reusable organizational cost knowledge base

Example of Cost Estimate

Surveyor Pro Project Cost Estimate Created October 5					
	# Units/Hrs.	Cost/Unit/Hr.	Subtotals	WBS Level 2 Totals	% of Total
WBS Items					
1. Project Management				\$306,300	20%
Project manager	960	\$100	\$96,000		
Project team members	1,920	\$75	\$144,000		
Contractors (10% of software development and testing)			\$66,300		
2. Hardware				\$76,000	5%
2.1 Handheld devices	100	\$600	\$60,000		
2.2 Servers	4	\$4,000	\$16,000		
3. Software				\$614,000	40%
3.1 Licensed software	100	\$200	\$20,000		
3.2 Software development*			\$594,000		
4. Testing (10% of total hardware and software costs)			\$69,000	\$69,000	5%
5. Training and Support				\$202,400	13%
Trainee cost	100	\$500	\$50,000		
Travel cost	12	\$700	\$8,400		
Project team members	1,920	\$75	\$144,000		
Subtotal			\$1,267,700		
6. Reserves (20% of total estimate)			\$253,540	\$253,540	17%
Total project cost estimate				\$1,521,240	

*See software development estimate.

FIGURE 7-2 Surveyor Pro project cost estimate

3. Determining Budget

- It is the process of aggregating the estimated costs of individual activities or work packages to establish an **authorised cost baseline**.
- This step produces the total project budget → used to monitor and control costs.
- Budget = Cost Estimates + Contingency Reserves + Management Reserves.

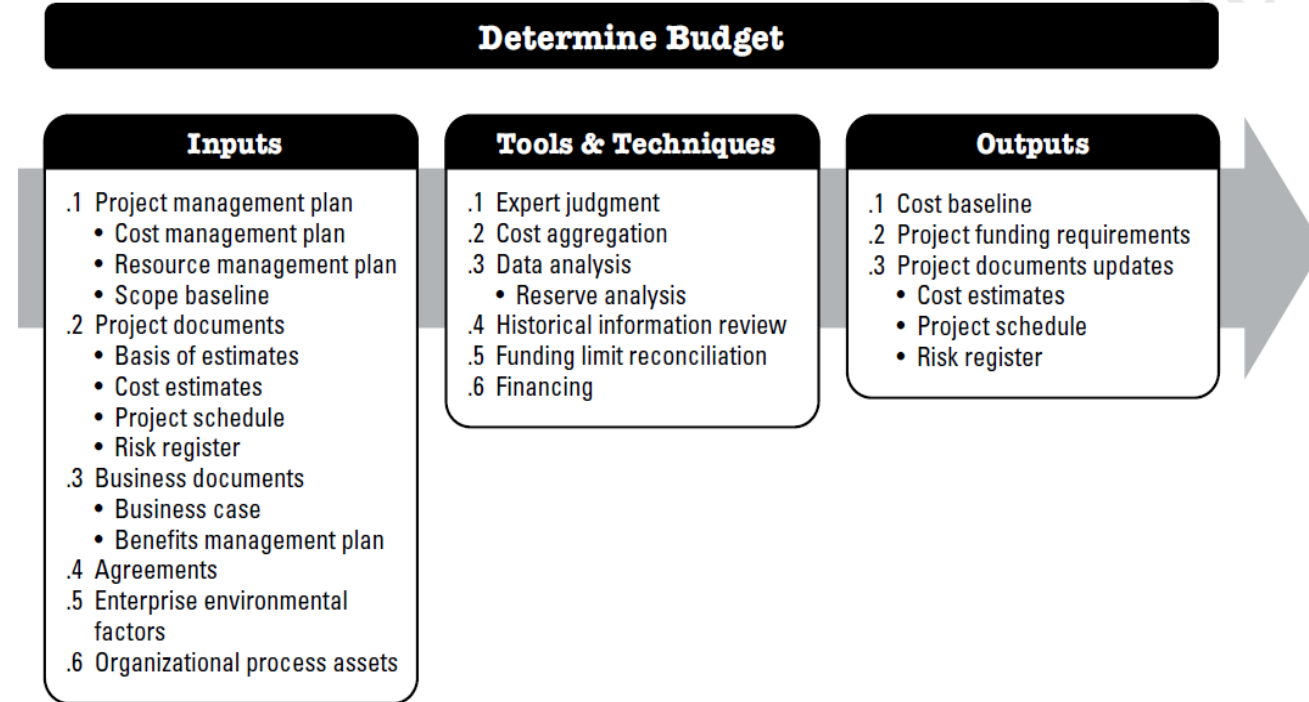


Figure 7-6. Determine Budget: Inputs, Tools & Techniques, and Outputs

Determining the Budget

- The main purpose of this step is:
 - Provides the spending plan for the project.
 - Ensures project has funding authorization.
 - Establishes the time-phased cost baseline for tracking (Planned Value, PV).
 - Defines reserves for risks and unknowns.
 - Communicates financial expectations to stakeholders.
- The main components of a Project Budget is:
 - Direct Costs: salaries, hardware, software, vendor fees.
 - Indirect Costs: overhead, admin, office space.
 - Contingency Reserves: buffer for identified risks.
 - Management Reserves: buffer for unknown risks.
 - Cost Baseline: approved budget (without management reserves).
 - Total Project Budget = Cost Baseline + Management Reserves.

Cost Baseline

- It is the approved, **time-phased budget** for the project.
- It is developed by summing up **estimated costs** by work packages and scheduling them over time.
- Used as the **benchmark** against which actual performance is measured.
- It includes the direct costs, indirect costs, and contingency reserves.

Example of Cost Baseline

Surveyor Pro Project Cost Baseline Created October 10*

WBS Items	Months												Totals
	1	2	3	4	5	6	7	8	9	10	11	12	
1. Project Management													
1.1 Project manager	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	96,000
1.2 Project team members	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	144,000
1.3 Contractors		6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	66,300
2. Hardware													
2.1 Handheld devices				30,000	30,000								60,000
2.2 Servers				8,000	8,000								16,000
3. Software													
3.1 Licensed software				10,000	10,000								20,000
3.2 Software development		60,000	60,000	80,000	127,000	127,000	90,000	50,000					594,000
4. Testing			6,000	8,000	12,000	15,000	15,000	13,000					69,000
5. Training and Support													
5.1 Trainee cost									50,000				50,000
5.2 Travel cost									8,400				8,400
5.3 Project team members							24,000	24,000	24,000	24,000	24,000	24,000	144,000
6. Reserves				10,000	10,000	30,000	30,000	60,000	40,000	40,000	30,000	3,540	253,540
Totals	20,000	86,027	92,027	172,027	223,027	198,027	185,027	173,027	148,427	90,027	80,027	53,567	1,521,240

*See the lecture slides for this chapter on the Instructor website for a larger view of this and other figures in this chapter. Numbers are rounded, so some totals appear to be off.

FIGURE 7-4 Surveyor Pro project cost baseline

4. Controlling costs

- It is the process of monitoring the status of the project to update the project costs and managing changes to the cost baseline.
- The key purpose of this step is to:
 - Ensure spending is within the approved budget.
 - Detect variances early.
 - Take corrective actions before overruns escalate.

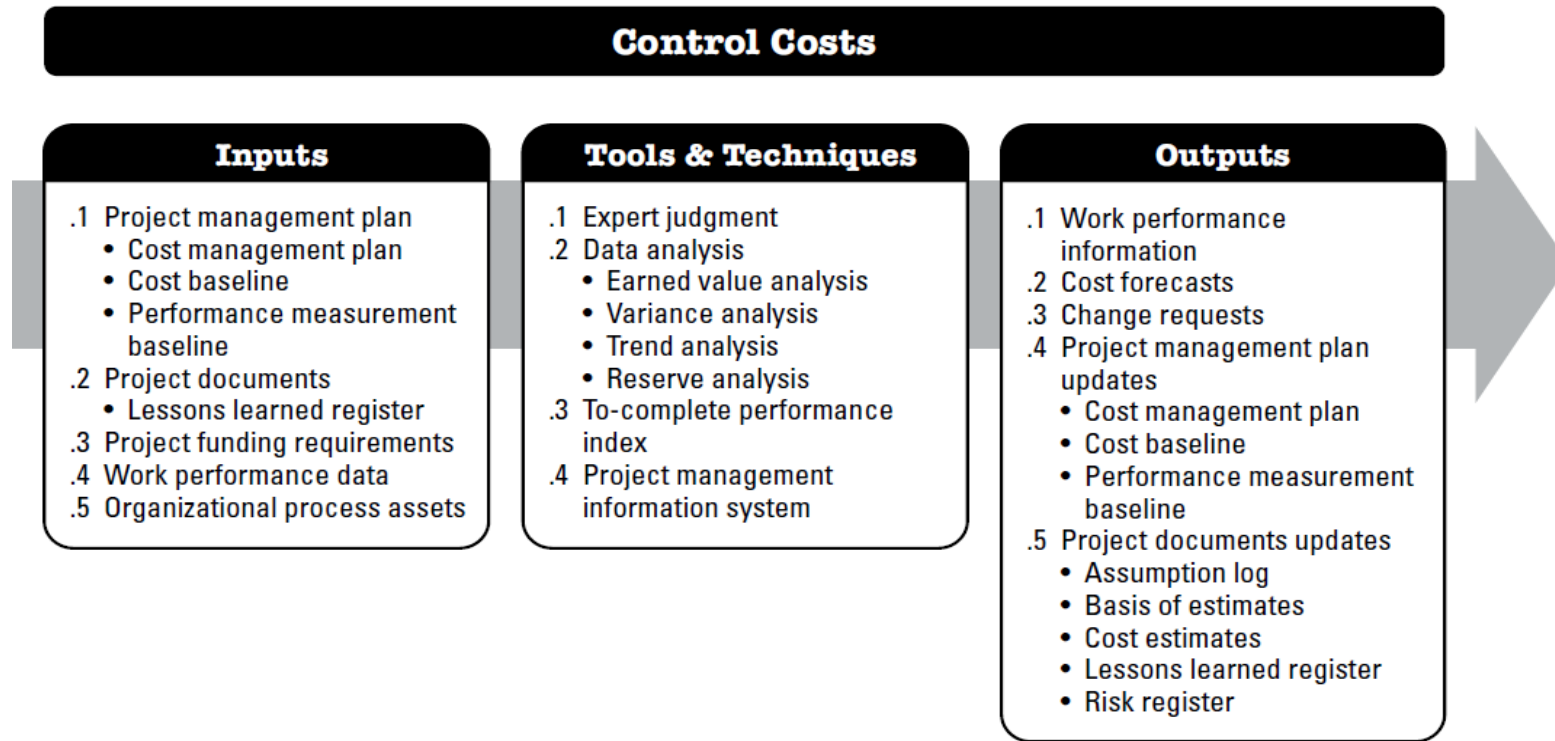


Figure 7-10. Control Costs: Inputs, Tools & Techniques, and Outputs

Earned Value Management – A tool of controlling costs

- EVM is a methodology that integrates **scope, schedule, and cost** to objectively measure project performance and progress.
- It answers 3 key questions:
 - Where are we now in terms of performance to date?
 - Where are we going? (forecast)
 - Will we finish on budget and schedule?
- Benefits of EVM:
 - Provides early warning signs of overruns.
 - Enables objective, data-driven reporting.
 - Integrates scope, cost, and schedule → holistic view.

Core components of EVM

- Planned Value (PV):
 - Budgeted cost of work planned to be done by a certain date.
 - The total cost estimate planned to be spent on an activity during a given period
- Earned Value (EV):
 - Budgeted cost of work actually completed by a certain date.
 - The estimate of the value of the physical work actually completed
- Actual Cost (AC):
 - Real expenditure for the work performed.
 - The total direct and indirect costs incurred in accomplishing work on an activity during a given period.

EVM - Variance Analysis

- Cost Variance (CV):
 - The amount of budget deficit or surplus at a given point of time.
 - It is the difference between Earned Value and Actual Cost
 - $CV = EV - AC$
 - If CV is Positive → under budget & Negative → over budget
- Schedule Variance (SV):
 - The measure of schedule performance.
 - It is the difference between the Earned Value and the Planned Value
 - $SV = EV - PV$
 - If SV is Positive → ahead of schedule & Negative → behind schedule

EVM – Performance Indexes

- Cost Performance Index (CPI):
 - A measure of Cost efficiency of budgeted resources
 - Expressed as ratio of Earned Value to Actual Cost
 - $CPI = EV/AC$
 - If $CPI > 1 \rightarrow$ cost efficient and if $CPI < 1 \rightarrow$ cost overrun
- Schedule Performance Index (SPI):
 - A measure of Schedule efficiency
 - Expressed as ratio of Earned Value to Planned Value
 - $SPI = EV/PV$
 - If $SPI > 1 \rightarrow$ ahead of schedule and if $SPI < 1 \rightarrow$ behind schedule

EVM – Forecasting numbers!

- Estimate at Completion (EAC):
 - The **expected total cost** of the project **based on current performance trends**.
 - It answers: *“Given how we’re performing now, how much will the project actually cost in the end?”*
 - EAC can be recalculated during the project when variances occur.
- Budget at Completion (BAC):
 - The **total planned budget** for the entire project.
 - It’s the sum of all planned costs (from the Cost Baseline).
 - Think of it as: *“How much we originally expected to spend when the project finishes.”*

Cost Management Approaches: Traditional vs Agile

- Traditional (Waterfall/Plan-driven)
 - Budget fixed upfront, cost baseline created early
 - Detailed estimation during planning (WBS, reserves, baselines)
 - Earned Value Management (EVM) aligns well: PV, EV, AC tracked consistently
 - Predictable, suited for compliance and contract-heavy projects
- Agile (Iterative/Adaptive)
 - Budgets allocated per iteration or release (timeboxed funding)
 - Estimates evolve using rolling-wave planning
 - EVM less straightforward: scope changes frequently, value measured in features or story points
 - Flexible, suited for dynamic environments with changing requirements

Challenges of Applying EVM in Agile Projects

- Challenges
 - Changing scope makes Planned Value unstable
 - Story points and features are not directly measurable in dollars
 - Frequent reprioritisation reduces the usefulness of variance analysis
 - Sponsors often expect cost forecasts in monetary terms rather than velocity metrics
- Adaptations
 - Track Earned Business Value (EBV) instead of only cost efficiency
 - Map story points to cost using sprint burn rates
 - Recalculate EAC at the end of each iteration using rolling forecasts

Best Practices for Agile Cost Management

- Fund increments (sprints or releases) instead of the entire project upfront
- Use burn-up or burn-down charts and velocity as proxies for value delivery
- Combine Agile tracking with lightweight EVM reporting for executives
- Maintain contingency reserves even in Agile projects
- Communicate clearly with stakeholders: Agile fixes time and cost but scope remains flexible

Traditional vs Agile Cost Management

Aspect	Traditional (Waterfall/Plan-driven)	Agile (Iterative/Adaptive)
Budgeting	Fixed upfront, detailed cost baseline	Incremental, funded per sprint/release
Estimation	Comprehensive planning, WBS-based	Rolling-wave, evolving estimates
Scope Stability	Scope fixed early, baseline stable	Scope flexible, backlog reprioritised
EVM Application	Works well: PV, EV, AC tracked reliably	Difficult: scope & metrics change frequently
Measurement of Value	Monetary units (dollars, hours, cost)	Story points, features, business value
Stakeholder reporting	Financial variance reports (CV, SV, CPI, SPI)	Velocity charts, burn-up/down, business value
Risk handling	Reserves set upfront (contingency, management)	Flexibility built into scope & incremental funding

Agile Budgeting in a Software Development Project

- Scenario
 - A company is developing a mobile banking app using Scrum. Instead of setting a fixed cost baseline for the entire project, the budget is allocated incrementally.
- Team setup
 - 1 Product Owner
 - 1 Scrum Master
 - 6 Developers
 - Average team cost: \$60,000 per sprint (2 weeks)

Agile Budgeting in a Software Development Project

- Budgeting Approach
 - Management approves funding for 10 sprints upfront = \$600,000 (time & cost fixed).
 - Scope is flexible, backlog items reprioritised each sprint to deliver maximum value.
 - At the end of Sprint 5 (halfway), stakeholders review progress:
 - Delivered: Login, payments, notifications features
 - Remaining budget: \$300,000
 - Decision: Add biometric login (new feature) by adjusting priorities without increasing budget.
- In Agile, time and cost are often fixed, but scope is variable. The team delivers the highest-value features first, adjusting scope to stay within the allocated budget.

Earned Value Management Formulae

Term	Short Definition	Formula
Planned Value (PV)	Value of the work planned to be completed by a certain time	—
Earned Value (EV)	Value of the work actually completed	$EV = PV \text{ of completed work}$
Actual Cost (AC)	Actual money spent to complete the work	—
Cost Variance (CV)	Difference between earned value and actual cost (budget performance)	$CV = EV - AC$
Schedule Variance (SV)	Difference between earned value and planned value (schedule performance)	$SV = EV - PV$
Cost Performance Index (CPI)	Measures cost efficiency of the project	$CPI = EV / AC$
Schedule Performance Index (SPI)	Measures schedule efficiency of the project	$SPI = EV / PV$
Estimate at Completion (EAC)	Estimated total project cost when the project finishes	$EAC = BAC / CPI$
Estimate to Complete (ETC)	Estimated cost required to finish remaining work	$ETC = EAC - AC$

Example for EVM Calculation

- Your budgeted cost of an IT project is \$100,000 and it needs to be completed in 6 months. After 2 months of completion, you (PM), wish to assess the project performance and progress. You have found that your team has completed 30% of the project work and spent \$40,000. At this stage, your planned completion is 35%. How is your project performing?

Step 1: Calculate Key Values

Metric	Calculation	Result
Earned Value (EV)	$30\% \times 100,000$	\$30,000
Planned Value (PV)	$35\% \times 100,000$	\$35,000
Actual Cost (AC)	Given	\$40,000

Example for EVM Calculation cont.

Step 2: Project Performance

Metric	Formula	Calculation	Meaning
Cost Variance (CV)	$EV - AC$	$30,000 - 40,000$	-10,000; Over budget
Schedule Variance (SV)	$EV - PV$	$30,000 - 35,000$	-5,000; Behind schedule
Cost Performance Index (CPI)	EV / AC	$30,000 / 40,000 = 0.75$	Over budget (as $0.75 < 1$)
Schedule Performance Index (SPI)	EV / PV	$30,000 / 35,000 = 0.857$	Behind schedule (as $0.85 < 1$)

Example for EVM Calculation cont.

Step 3: Forecasting

Metric	Formula	Calculation	Meaning
Estimate at Completion (EAC)	BAC / CPI	$100,000 / 0.75$	\$133,333 expected final cost
Variance at Completion (VAC)	$BAC - EAC$	$100,000 - 133,333$	-\$33,333 over budget

Challenges in IT Cost Management

- Scope creep and requirement volatility
 - IT projects often face **changing requirements** (new features, security upgrades, compliance changes). This leads to additional costs that weren't part of the original budget.
- Hidden and recurring costs
 - Licenses, cloud subscriptions, SaaS fees, and vendor support may be overlooked. Cloud costs are **consumption-based** → unpredictable if not monitored.
- Measurement challenges
 - Difficult to track **actual progress** vs. costs spent.
- Vendor and outsourcing risks
 - Many IT projects outsource components. This causes the existence of a lot of risks such as vendor delays, hidden costs in SLAs, currency fluctuations for offshore teams, etc.

Industry Example of IT Cost Blowouts

- ERP Implementation – Hershey’s (1999)
 - Budget: \$112M
 - Issues: Overly ambitious timeline, poor testing, inadequate training
 - Outcome: \$100M order backlog, stock dropped 8% → project seen as a failure despite completion (*O’Leary, 2000; CIO Magazine*)
- Cloud Migration – US Army (2018)
 - Initial Budget: \$550M
 - Issues: Hidden costs in vendor contracts, underestimated integration, cybersecurity compliance costs
 - Outcome: Budget ballooned to \$1.6B before corrective action (*GAO Report, 2019*)

Industry Example of IT Cost Blowouts

- Healthcare.gov (2013)
 - Budget: \$93M
 - Issues: Scope creep, vendor coordination failures, underestimated testing costs
 - Outcome: Final cost exceeded \$292M, rollout delays damaged stakeholder trust (*U.S. Government Accountability Office, 2014*)
- Sydney Trains Digital Systems (2018–ongoing)
 - Original Budget: \$880M
 - Issues: Complex vendor ecosystem, underestimated testing/integration costs, evolving regulatory requirements
 - Outcome: Revised budget \$1.1B+, with further delays expected into late 2020s (*NSW Audit Office, 2021*)

Best Practices

- Start with a strong cost management plan
 - Define **how costs will be planned, estimated, budgeted, and controlled.**
- Use a detailed and realistic work breakdown structure
 - Break deliverables down into manageable work packages. Assign **costs at the activity/work package level** for accuracy.
- Build in contingency and management reserves
 - Use **past project data** to improve estimates. Benchmark against industry averages (e.g., cost per user migrated to cloud). Furthermore, create an **organizational cost knowledge base** for future projects.
- Monitor costs continuously and not just at milestones
 - Use **real-time dashboards** (Power BI, Tableau, MS Project). Track **Planned Value vs. Earned Value vs. Actual Cost**. Spot variances early before they snowball.

References

- Schwalbe, K., 2018. Project Schedule Management. In: *Information Technology Project Management*. Boston: Cengage, pp. 285-326.

For You!

End of Lecture Questions

- What is the purpose of a **Project Cost Management Plan**?
- Define **Planned Value (PV)**, **Earned Value (EV)**, and **Actual Cost (AC)**.
- What is a **Cost Baseline**, and why is it important in project management?
- Explain the difference between **contingency reserves** and **management reserves**.
- An IT project team is using Agile methodology. What challenges might arise in applying **EVM** to this environment, and how could you adapt the approach?

Case Study

- The University of Sydney plans to implement a **new online student portal** to manage course registration, exam scheduling, and student records. The project is expected to last **6 months**, with a total budget of **\$180,000**.
- Scope highlights:
 - Online course registration
 - Exam scheduling module
 - Student records and reporting
 - Integration with existing LMS
 - Training for staff

Case Study

- Planned Schedule and Costs

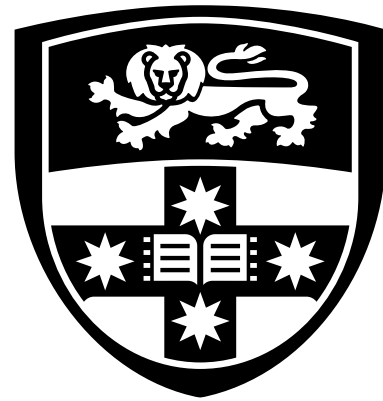
Month	Planned Cost (PV)
1	\$20,000
2	\$35,000
3	\$30,000
4	\$35,000
5	\$30,000
6	\$30,000
Total	\$180,000

- Additional Information:

- Contingency Reserve: 10%
- Management Reserve: 5%
- Current progress at end of Month 3:
- Work completed: Course registration module fully done, exam scheduling 50% complete
- EV (Earned Value): \$80,000
- AC (Actual Cost): \$85,000

Case study based Questions

- Identify **key cost components** for this project.
- What **inputs** would you use to estimate the costs of the student portal project?
- Calculate the **Cost Baseline** including the contingency reserve.
- Explain how the cost baseline will help the project manager control costs.
- Using the information at the end of Month 3:
 - $PV = \$85,000$ (sum of first 3 months' planned costs)
 - $EV = \$80,000$
 - $AC = \$85,000$
- Calculate: CV, SV, SPI, CPI, and EAC assuming CPI will continue.
- Interpret the results of EVM.



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